Generative Retrieval-Augmented Hybrid-Attention Waveform Inference Machine

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Abstract

Traditional energy management systems often rely on predefined, logic-based algorithms to *improve system efficiency*. Although many of these systems incorporate forecasting techniques and are effective at balancing electrical loads, they may struggle to produce fine-grained, context-aware predictions of future electricity demand, particularly at scale. I introduce Generative Retrieval-Augmented Hybrid-Attention Waveform Inference Machine (GRaHWIM). the goal is making the energy *efficient*. GRaHWIM works by extrapolating the future need for electricity based on past usage pattern interpreted in the form of a graph(see *Figure 2*). To improve prediction stability, GRaHWIM performs N_x inference runs simultaneously and computes the mean of their predicted waveforms to produce a single aggregated forecast. Using the *optimal control theory* we could define "efficiency" as either *conserving as much energy as possible*, or *consuming all produced energy made by a source.* By using GRaHWIM, we could reduce wasted energy from a source to **almost none**.

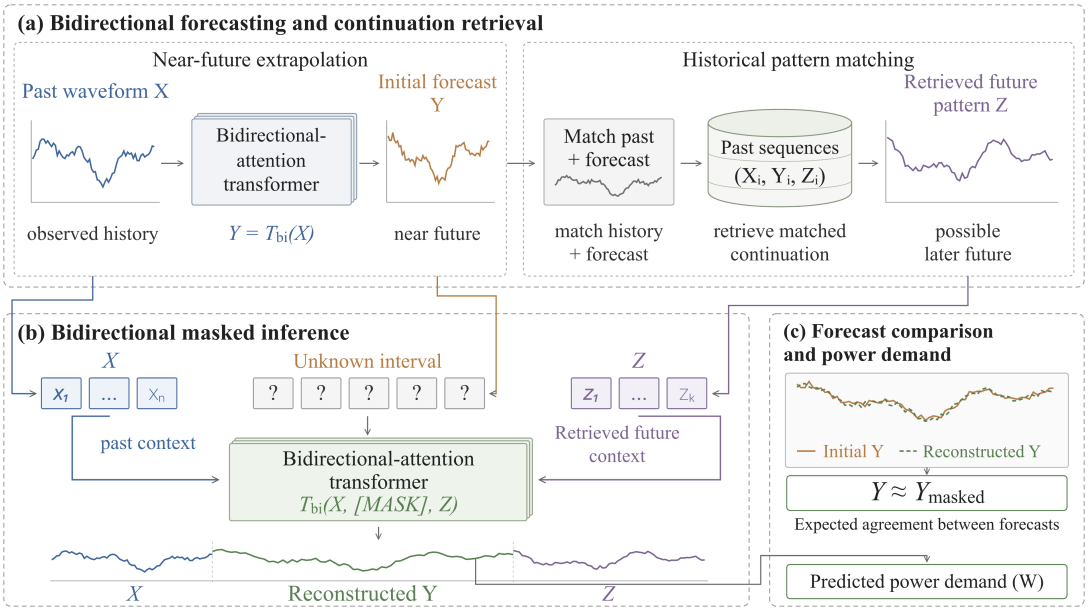


Figure 1: how GRaHWIM works in a high level

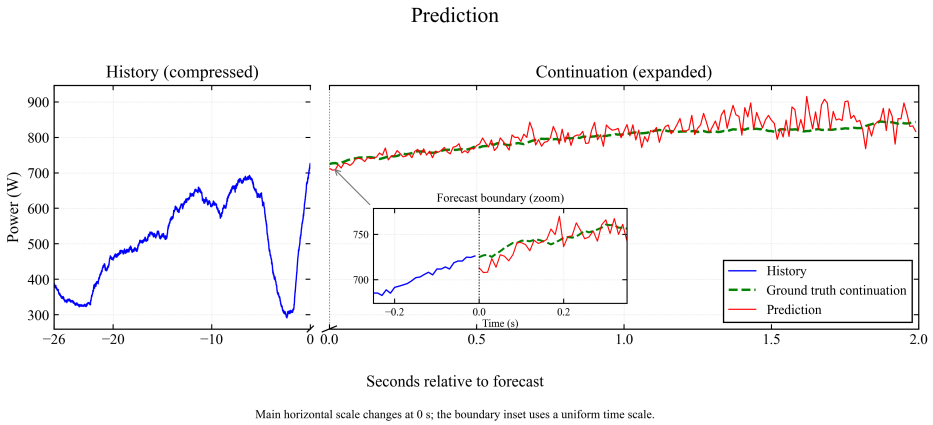


Figure 2: Example of an appliance power consumption waveform

1 Introduction

The efficient management of electrical energy is an important challenge in modern power systems, particularly in environments where electricity generation is limited or varies over time. Electrical appliances exhibit different consumption patterns depending on their operating states, environmental conditions, and user interactions. Consequently, the demand for electricity can fluctuate significantly over short periods of time. Maintaining an appropriate balance between available generation and electrical demand is therefore essential for improving energy utilization, minimizing unnecessary consumption, and ensuring the reliable operation of connected components.

Traditional energy management systems commonly employ predefined control rules, load prioritization strategies, and threshold-based algorithms to regulate electricity distribution. More advanced systems incorporate forecasting techniques to anticipate changes in electrical demand and adjust their allocation strategies accordingly. Although these methods can be effective, accurately predicting short-term, appliance-level electricity consumption remains challenging, particularly when operating conditions change unexpectedly. In addition, predictions based on historical consumption patterns may not adequately represent the complex and nonlinear behavior of individual electrical components.

A significant challenge in electricity management is determining how much power an individual component will require at a particular point in the near future. While reactive control strategies respond to changes in measured demand, predictive approaches can potentially identify upcoming changes before they occur. Such information may allow an energy management system to prepare for anticipated increases in consumption, prioritize essential electrical loads, and allocate available generation more effectively. However, the usefulness of these predictions depends on their accuracy, computational efficiency, and ability to adapt to previously unseen consumption patterns.

To investigate this problem, I introduce the *Generative Retrieval-Augmented Hybrid-Attention Waveform Inference Machine* (GRaHWIM), a proposed framework for predicting short-term electrical consumption through the analysis of historical power waveforms. GRaHWIM represents electrical consumption as time-dependent numerical sequences, allowing its forecasting models to identify temporal patterns and extrapolate possible future demand. Rather than relying exclusively on a single prediction, the framework performs multiple inference runs and aggregates their predicted waveforms to produce a consolidated forecast. This approach is intended to improve prediction stability and reduce the influence of variability between individual inference results.

An additional component of GRaHWIM is its retrieval-augmented forecasting mechanism. Historical consumption waveforms and previously generated inference records can be stored and subsequently retrieved when similar operating conditions are encountered. By comparing newly observed consumption patterns with previously recorded examples, the system aims to incorporate relevant historical information into its forecasting process. This mechanism is designed to investigate whether the continuous accumulation and retrieval of historical records can improve predictive performance over time, particularly when the system encounters recurring appliance behaviors.

Beyond forecasting electricity consumption, GRaHWIM is intended to support more effective energy allocation decisions. Using principles from *optimal control theory*, the system aims to evaluate how predicted electrical demand can be incorporated into decisions that balance energy conservation, available generation, and the operational requirements of connected components. Depending on the constraints and objectives of the electrical system, this may involve reducing unnecessary consumption, minimizing unused available generation, or prioritizing electrical loads according to their anticipated requirements. The effectiveness of such decisions depends not only on prediction accuracy but also on the physical limitations and operational constraints of the energy source and connected appliances.

This research investigates the feasibility of combining generative waveform forecasting, historical retrieval, and predictive energy allocation within a unified framework. A synthetic appliance simulation environment is used to evaluate the proposed approach under varying consumption patterns and operating conditions. The evaluation considers forecasting accuracy, computational requirements, and the potential effects of prediction-informed allocation strategies. Through these experiments, the research aims to determine whether GRaHWIM provides a practical foundation for adaptive, fine-grained electrical energy management.

2 Problem Definition and Objectives

I consider an electrical system in which several appliances request power from a source with limited or changing output. The first problem is to estimate the next part of an appliance's power trace from observations that have already arrived. The second problem is to decide how much of the available power should be allocated to each appliance and when a flexible task should run.

Let $p_i[k]$ be the actual demand of appliance i at sample k, and $x_i[k]$ its noisy observation. At a forecast origin k, the model estimates

$$\hat{\mathbf{p}}_i = f\big(x_i[k-L+1:k]\big),$$

where L is the number of observed samples. The output contains H future power readings. For my fan experiment, $L = 260$ and $H = 40$ at 10 Hz, corresponding to 26 seconds of history and four seconds of future demand.

Here, a *waveform* is a trace of active power in watts. Energy is the integral of power over time. If power is in watts and Δt is in seconds, the energy in watt-hours over H samples is

$$E_{\mathrm{Wh}} = \frac{\Delta t}{3600} \sum_{h=1}^{H} p_i[k+h].$$

This distinction matters for the objective. A forecast with lower error can improve an allocation decision. The allocation itself must still be compared with a baseline to determine what energy or grid purchases it actually avoids.

2.1 Objectives

I evaluate whether the forecast improves on repeating the last observed power level, and whether it follows a rise or fall rather than returning a constant estimate. For the control stage, I separately examine how delaying a flexible task can reduce grid purchases without reducing the required service. The experiments evaluate simulated demand and a separate scheduling scenario. The aspiration in the abstract concerns a future controller, and the reported results apply to the specific conditions described below.

3 System Architecture

Figure 1 shows the architecture I propose. I separate the system into observation, prediction, retrieval and allocation so each stage can be evaluated independently. The forecasting experiment uses the observation and prediction stages. The retrieval and live allocation stages describe the intended integrated design.

3.1 Observation and prediction

The observation stage maintains a rolling history of power for each appliance. The predictor receives this history and produces a sequence of future power estimates. I evaluate autoregressive text generation with LFM3, a numerical prediction head attached to a bidirectional encoder, and a smaller model trained directly on the fan's numerical signal. The bidirectional encoder is based on ModernBERT [1]. In the continuation task, all future positions are masked. A separate gap reconstruction task could use observations on both sides of an interval that has already elapsed.

The architecture allows multiple forecasts to be averaged as described in the abstract. The benchmark below evaluates one deterministic prediction per model, so it does not measure an advantage from running multiple samples. When two predictions place a change at different times, averaging them can soften the change. I therefore score both the individual power readings and the direction of the change.

3.2 Retrieval and control

The retrieval stage would index observed prefixes and their associated continuations from earlier independent runs. I would retrieve records from the same appliance type using distances over both shape and power level. The idea is related to retrieval-augmented generation [2], with an appliance trace in place of a text document. Only the observed query prefix can be used when finding a match. A candidate from the same episode or the test set would be excluded. If the match is useful, the controller could blend the aligned continuation p_{ref} with the model estimate p_{mod}:

$$\hat{p} = \alpha p_{\mathrm{mod}} + (1-\alpha) p_{\mathrm{ref}}.$$

The coefficient α would be selected on validation episodes. The intended allocation stage receives the forecast, appliance priorities and a source power budget $S[k]$. Its proposed budgets $a_i[k]$ and reserve $r[k]$ must satisfy

$$\sum_i a_i[k] + r[k] \le S[k].$$

The reserve represents instantaneous capacity held back for unexpected demand. Allocation must

also account for minimum service and the deadline of every flexible task. This use of a forecast to make decisions under constraints follows the general structure of model predictive control [3].

4 Methodology

4.1 Data generation

I generate synthetic active power at 100 Hz. The fan's request changes between operating regimes, its speed responds with inertia, and its demand grows approximately with the cube of speed. The simulated active range is 5–80 W. The generator adds correlated operating variation and simulated measurement noise. I average blocks of ten observations for model input and score the clean simulated demand at 10 Hz. The prediction models see only the observed prefix, with the simulator seed and future operating conditions withheld.

4.2 Numerical forecasting model

For the small fan model, I use one average per second for the earlier 20 seconds and ten averages per second for the final six seconds. This keeps the recent changes visible while reducing the number of inputs. The model uses two hidden layers of 96 units each and predicts nine knots across the four-second horizon. I linearly interpolate between the knots to obtain the 40 forecast readings. The outputs are changes relative to the mean of the last half-second of observed power.

I train with a smooth absolute error objective on both the readings and their one-second changes. For $e_h = \hat{p}_h - p_h$, the objective has the form

$$\mathcal{L} = \frac{1}{40}\sum_{h=1}^{40}\rho(e_h) + \frac{0.3}{30}\sum_{h=11}^{40}\rho(e_h - e_{h-10}),$$

where ρ is the smooth L1 loss. In training, the errors in this expression are scaled by 10 W, and the loss switches from quadratic to linear at a scaled magnitude of 0.2.

4.3 Evaluation protocol

I generated 800 training episodes and extracted 31 windows from each, yielding 24,800 training windows. Another 96 episodes were used for validation. I evaluated the models on 24 shared episodes, then evaluated the numerical model and persistence on a separate set of 96 episodes. I assigned whole episodes to a split before extracting windows. Normalization was fitted using training observations.

I use persistence, defined as the mean observed power in the last second repeated across the horizon, as the baseline [4]. I report mean absolute error (MAE) and the proportion of points within 5 W of the actual future. A forecast is nearly flat if its range is at most 1 W. For direction accuracy, I compare the sign of the difference between the last and first half-second means, on episodes where the actual difference is at least 5 W in magnitude.

5 Results

5.1 Fan demand forecasting

Table 1: Forecasts on the same 24 simulated fan episodes. Direction is evaluated on 13 episodes with a clear rise or fall.

Model	MAE (W)	Within 5 W	Direction
Numerical fan	4.51	72.8%	10/13
LFM3 (12k)	4.75	69.7%	3/13
Persistence	5.78	63.5%	0/13

The numerical model produced nearly flat forecasts in 4 of the 24 episodes, compared with 22 for LFM3. Its mean error was only 0.24 W lower than LFM3, but it captured the direction of more of the clear rises and falls. The fan ModernBERT model averaged 13.69 W MAE on the same episodes. Its forecast varied by less than 1 W after binning on all 24 episodes.

On the separate 96 episodes, the numerical model averaged 4.37 W MAE, compared with 6.95 W for persistence. It got the direction right on 33 of 40 clear changes. Its first two seconds averaged 2.26 W error. Across the 53 episodes whose actual power varied by more than 5 W, the model averaged 6.93 W error, compared with 11.25 W for persistence. Abrupt changes that begin after the forecast origin can still produce substantial misses.

5.2 Illustrative school scheduling study

I consider a school wing from 08:00 to 16:00 with essential demand of 2.4 kW before 09:00, 3.0 kW from 09:00 to 15:00, and 2.6 kW afterward. A 1.2 kW flexible task must run for two consecutive hours. The assumed solar profile is a half sine with a nominal 5 kW peak. With 15-minute bins, imported energy is

$$E_{\text{grid}} = 0.25\sum_{j}\max(P_j - S_j, 0),$$

where P_j and S_j are load and solar power in kW, giving E_{grid} in kWh. The scheduler evaluates feasible start times using the assumed clear-day profile and selects 10:30–12:30. The baseline starts at 08:00.

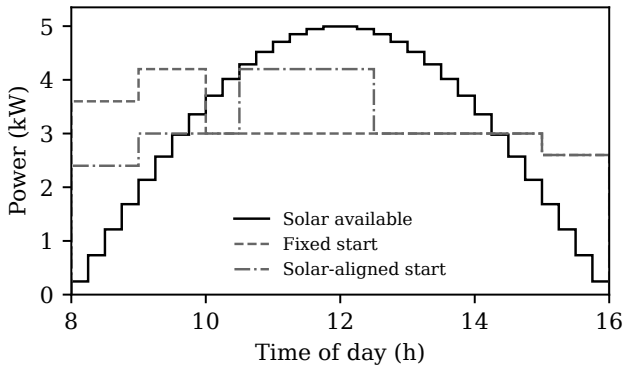


Figure 3: Assumed clear-day solar profile and simulated demand under two schedules. Grid imports occur where demand exceeds available solar power.

Table 2: Simulated grid imports in kWh, using the same two start times under each solar profile.

Solar profile	Fixed	Scheduled	Difference
Clear	6.03	3.72	2.31
Half output	12.66	12.66	0.00
Midday cloud	8.62	8.11	0.51

The clear-profile schedule reduces grid purchases by 2.31 kWh and increases on-site use of solar energy by the same amount. Total appliance consumption remains 25.40 kWh. At half solar output, demand exceeds generation throughout the day, so changing the start time offers no benefit. The cloud case assumes output drops to 35% of the clear profile from 11:00 to 13:00. The scheduling calculation uses these assumed profiles independently of the fan forecast.

5.3 Avoiding idle consumption

A school can also reduce total electricity use by switching off avoidable standby loads. If 20 workstation groups each have 3 W of switchable idle draw for 12 hours on 180 school days, the gross avoided consumption is 129.60 kWh. A shared controller drawing 2 W continuously would consume 17.52 kWh per year, leaving 112.08 kWh of net savings under these assumptions. A school trial would first measure actual idle draw [5] and account for existing sleep policies and the overhead of each switch.

6 Conclusion

I introduced GraHWIM to investigate whether an appliance's recent power trace could inform how electricity is allocated. The numerical model improved on persistence for the simulated fan and was more responsive to changing demand than the tested LFM3 checkpoint. The forecast remained less accurate on sudden changes, which matters when essential loads depend on the allocation.

The separate school scenario showed how the timing of a flexible task affects grid purchases. The task used the same energy in each schedule. Its benefit came from running during available solar generation. The standby example illustrates a different mechanism, in which switching off idle equipment reduces consumption after controller overhead is counted. Both examples depend on their stated inputs.

The next experiment would add retrieval and reserve sizing, then compare the resulting controller with a reactive baseline under identical generation and load traces. I would record grid imports, total consumption, unfinished tasks, unmet essential demand and decision latency. Measurements from actual appliances are necessary to find out whether the forecast transfers to a school and whether it yields a repeatable reduction in electricity use.

References

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